CFD SIMULATION OF TESLA MICROVALVES IN SERIES FOR MICROFLUIDIC FLOW RECTIFICATION

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Context and objective

Tesla microvalves rectify flow without moving parts. In the forward direction, the main channel provides lower resistance. In reverse flow, the branches redirect the fluid, promote recirculation, and increase pressure loss.

This study compares the pressure and velocity fields in both directions and calculates the diodicity of a multistage series configuration.

Methodology

The three-dimensional domain was generated in Gmsh and simulated in OpenFOAM. The simulations considered steady, incompressible, isothermal, and Newtonian flow using the `simpleFoam` solver and the SIMPLE algorithm.

Gmsh $\longrightarrow$ **OpenFOAM** $\longrightarrow$ **ParaView** $\longrightarrow$ **diodicity**

- Domain dimensions: $62.17 \times 12.75 \times 1.00$ mm.
- Nominal inlet channel width: 1.20 mm.
- Branch angle: $70°$.
- Mesh: 36,480 hexahedral cells.
- Compared cases: forward flow from the *inlet* to the *outlet*, and reverse flow from the *outlet* to the *inlet*.
- Post-processing in ParaView to extract pressure drops and inspect the velocity fields.

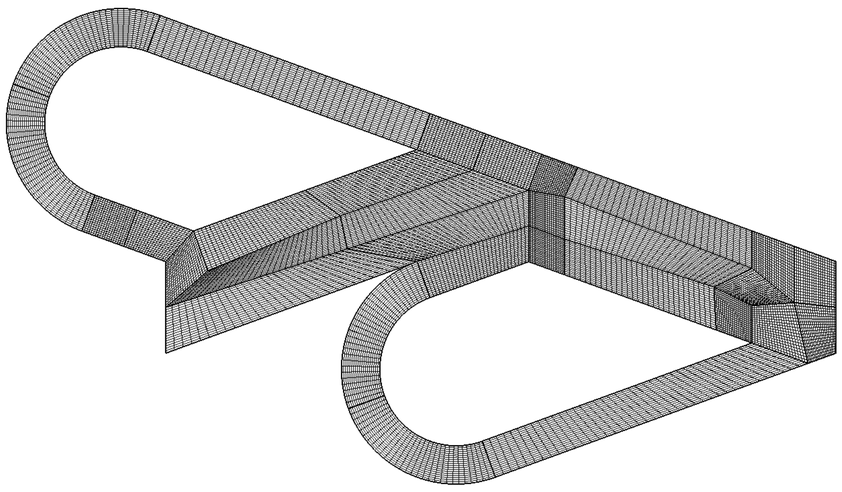


Figure 1. Detail of the three-dimensional mesh for one microvalve stage.

Performance metric

Diodicity compares pressure drops under equivalent conditions:

$$Di = \frac{\Delta P_{\mathrm{rev}}}{\Delta P_{\mathrm{dir}}}.$$

Values greater than 1 indicate higher hydraulic resistance in the reverse direction.

Conclusions and limitations

The pressure and velocity fields reveal the rectification mechanism. Forward flow follows the main channel. Reverse flow forces the fluid into the branches, produces recirculation, and increases pressure loss.

The series configuration reached $Di \approx 7.56$ under the simulated conditions. This value applies to this geometry and numerical setup. A mesh-independence study, an assessment of other Reynolds numbers, and comparison with experimental data are still required.

Essential references

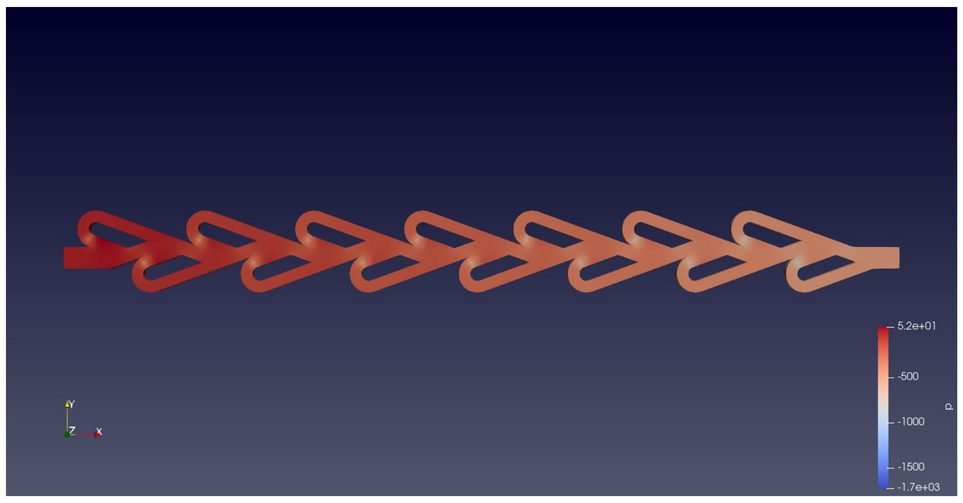
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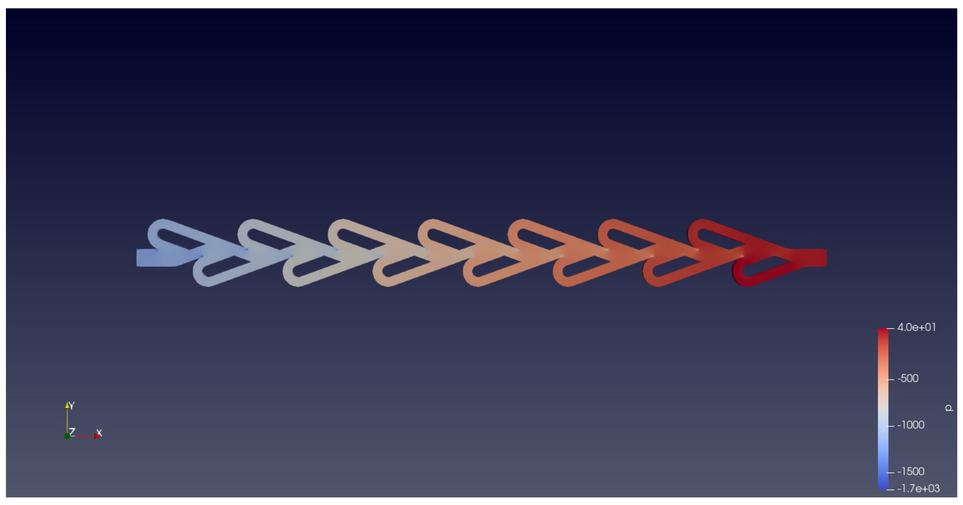
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Pressure fields

In forward flow, pressure decreases more gradually along the main channel. In reverse flow, the largest gradients occur at the bifurcations and return branches, where the geometry increases hydraulic resistance.



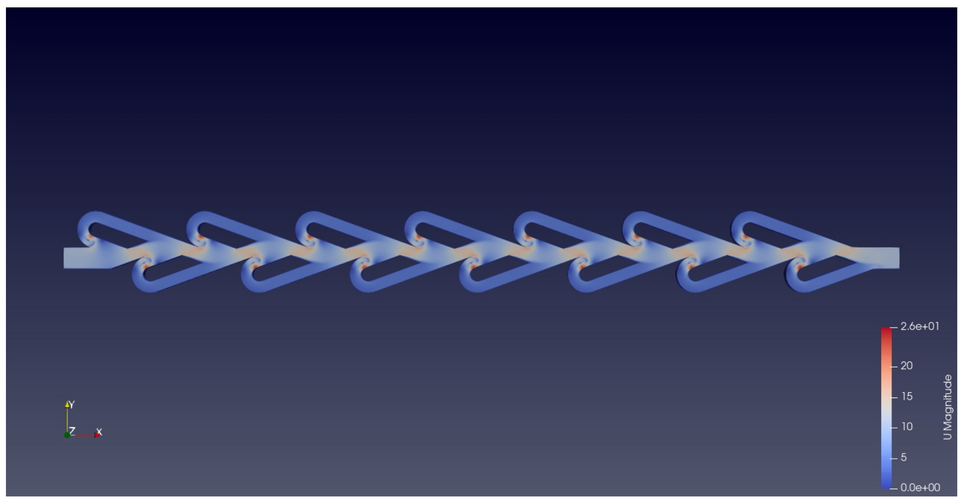
Forward. *inlet* $\rightarrow$ *outlet*.



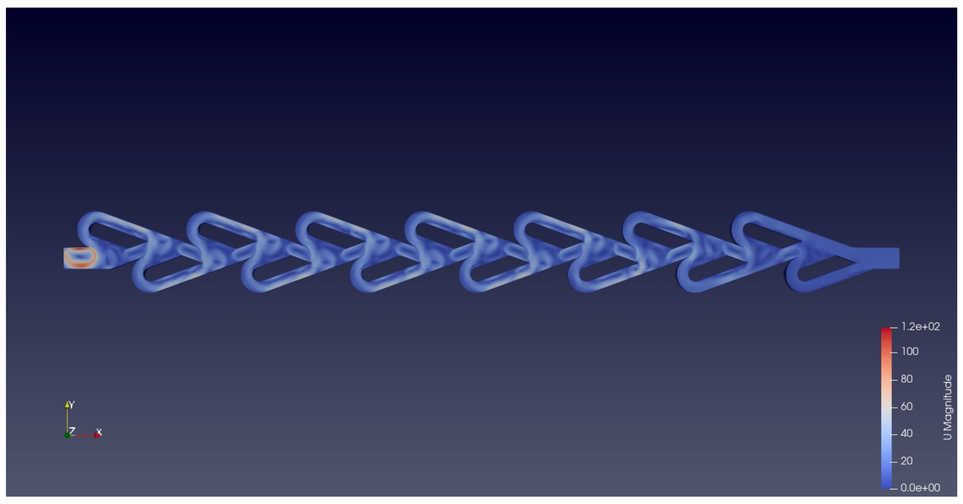
Reverse. *outlet* $\rightarrow$ *inlet*.

Velocity fields

In forward flow, the highest velocities remain concentrated in the main channel. In reverse flow, the fluid interacts with the branches and forms low-velocity and recirculation zones. This path dissipates more mechanical energy.



Forward. Limited branch engagement.



Reverse. Low-velocity and recirculation zones.

Main result

$$Di \approx 7.56$$

Direction	ΔP (Pa)	Relative resistance
Forward	0.00164095	lower
Reverse	0.0124002	higher

The reverse pressure drop was approximately 7.56 times the forward pressure drop.